December 16, 2025

Aix Marseille Université

Erasmus Mundus Master

Acoustic Imaging

Passive Imaging of a Scattering Cylinder using Ambient Noise

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1 Introduction

Conventional acoustic imaging, such as sonar or medical ultrasound, relies on "active" methods where a controlled source emits a pulse, and the system listens for echoes. However, in many real-world scenarios, we cannot control the source. This laboratory work explores **Passive Acoustic Imaging (PAM)**, a technique that exploits random, incoherent ambient noise to retrieve information about the environment.

The objective of this study is to reconstruct the image of a scattering object (an aluminum cylinder) placed inside a circular microphone array. Instead of a dedicated transmitter, we utilize a diffuse noise field generated by a moving source. The experiment proceeds in two stages:

1. **Green's Function Retrieval:** We demonstrate that the cross-correlation of diffuse noise recorded by two sensors allows us to estimate the impulse response (Green's function) between them.
2. **Beamforming:** We apply the Delay-and-Sum (Total Focusing Method) algorithm to these retrieved Green's functions to spatially locate the cylinder.

2 Theoretical Background

The core principle of this experiment is the fluctuation-dissipation theorem, applied to acoustics as the Lobkis-Weaver relation. It states that the cross-correlation of a diffuse field recorded at two points yields the Green's function between those points.

2.1 Noise Cross-Correlation

For two receivers at locations $\vec{r}_i$ and $\vec{r}_j$ recording time series $y_i(t)$ and $y_j(t)$, the cross-correlation is defined as:

$$\langle C_{i,j}(\tau)\rangle = \left\langle \int_{-\infty}^{+\infty} y_i(t) y_j(t+\tau)\, dt \right\rangle \tag{2.a}$$

To improve the signal-to-noise ratio and flatten the spectrum, the computation is performed in the frequency domain with energy normalization:

$$\langle C_{i,j}(\tau)\rangle = \mathcal{F}^{-1}\left(\frac{\tilde{y}_i(\omega)\tilde{y}_j^*(\omega)}{E_i E_j}\right) \tag{2.b}$$

2.2 Green's Function Retrieval

The time derivative of the noise cross-correlation approximates the Green's function $G(\vec{r}_i, \vec{r}_j, \tau)$:

$$\frac{d\langle C_{i,j}(\tau)\rangle}{dt} \approx -G(\vec{r}_i, \vec{r}_j, \tau) + G(\vec{r}_j, \vec{r}_i, -\tau) \tag{2.c}$$

This relation effectively turns one receiver into a virtual source and the other into a receiver, allowing us to perform imaging as if we had an active system.

3 Experimental Setup

The acquisition system consists of a circular array designed to encircle the target area.

- **Array:** 32 MEMS microphones arranged in pairs on a circular frame. The setup includes an inner ring and an outer ring.
- **Target:** An aluminum cylinder (radius $r \approx 4.5$ cm) placed inside the array.
- **Source:** A speaker emitting white noise (1 kHz - 10 kHz), manually moved around the setup to simulate a uniform diffuse field.

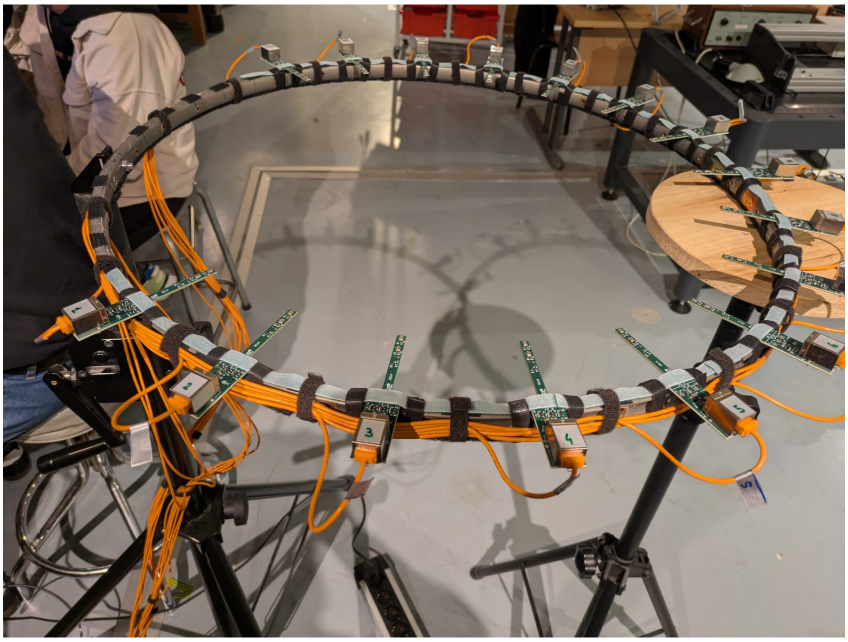

Figure 1: Experimental setup showing the circular array of MEMS microphones and the central target area.

Data was collected for two configurations: **Config 1** (With Cylinder) and **Config 2** (Empty Array). 50 independent acquisitions were recorded for each to ensure sufficient averaging.

4 Signal Processing & Results

4.1 Pre-processing and Filtering

The raw data was band-pass filtered between **1000 Hz and 9500 Hz** to eliminate low-frequency room modes and high-frequency aliasing. Following this, Welch's method was applied to compute the cross-correlation, using signal blocks of 1638 samples with 80% overlap.

Figure 2 displays the cross-correlation matrix relative to Reference Mic 1. The distinct "V-shape" or wavefront curvature indicates the time delays required for sound to propagate from Mic 1 to the other sensors across the array.

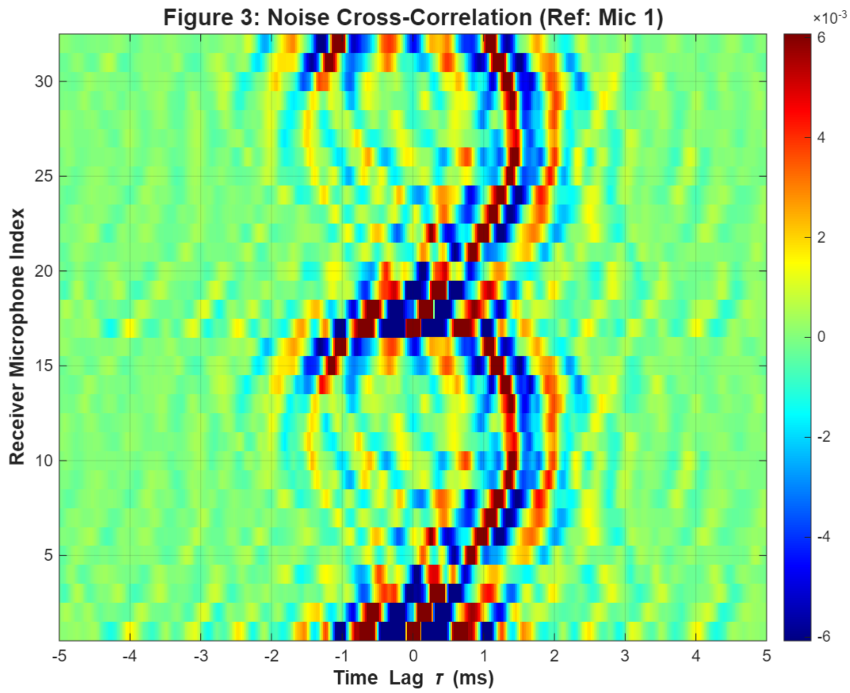

Figure 2: 2D Noise Cross-Correlation function between Reference Mic 1 and all other microphones. The clear wavefront pattern confirms the coherent propagation of sound extracted from noise.

4.2 Green's Function Retrieval

By taking the time derivative of the cross-correlation (Eq. 2.c), we retrieved the Green's Function. Figure 3 shows the result for the pair Mic 1 vs Mic 9 (located on opposite sides). The distinct peak at $\tau \approx 1.3$ ms corresponds to the direct time-of-flight across the array diameter, validating the physical reconstruction of the signal.

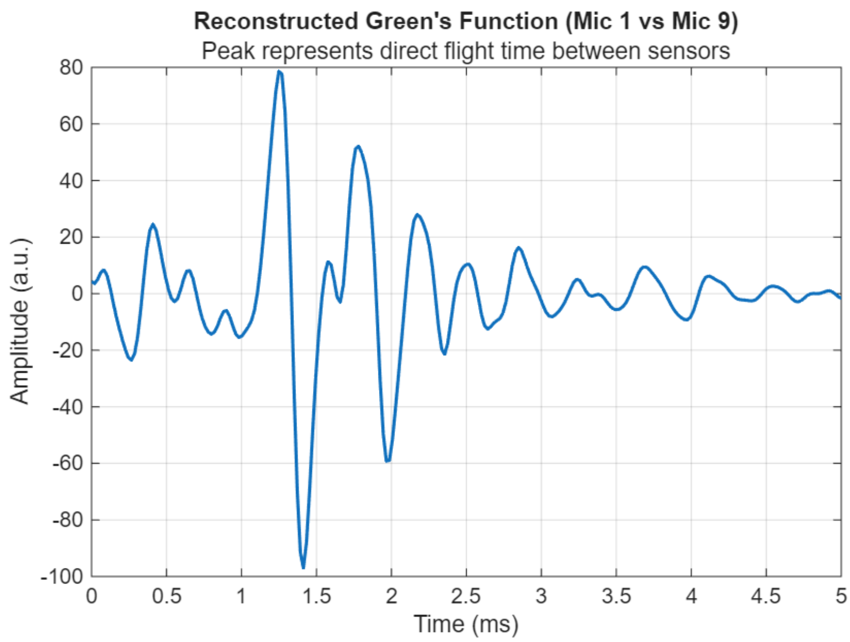

Figure 3: Reconstructed Green's Function between Mic 1 and Mic 9. The distinct pulse confirms the retrieval of the impulse response from ambient noise.

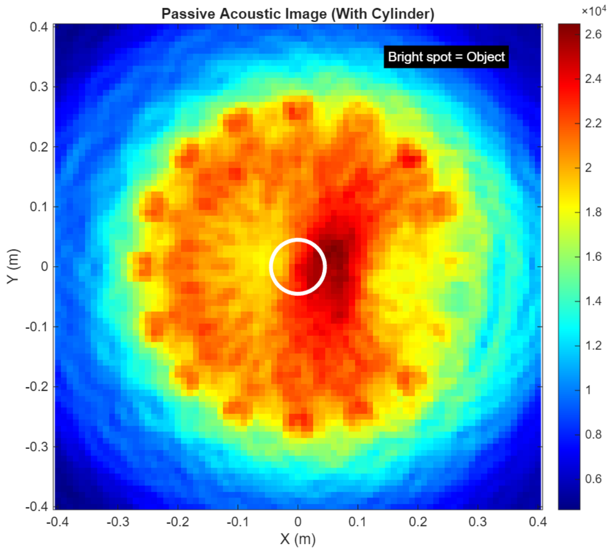

Figure 4: Positions of theme

5 Beamforming and Imaging

Using the Delay-and-Sum algorithm, we back-propagated the retrieved Green's functions to the imaging domain.

5.1 Detection Results

Config 1 (Cylinder) produced a strong scattering signature in the center of the domain. However, raw beamforming often contains artifacts from the direct path between microphones. To enhance the image quality, **Differential Imaging** was performed by subtracting the Empty configuration (Config 2) from the Cylinder configuration (Config 1):

$$I_{diff} = |I_{cylinder} - I_{empty}|$$

This process effectively removed the static background and the direct waves between sensors, leaving only the contribution of the scatterer. The result is shown in Figure 5.

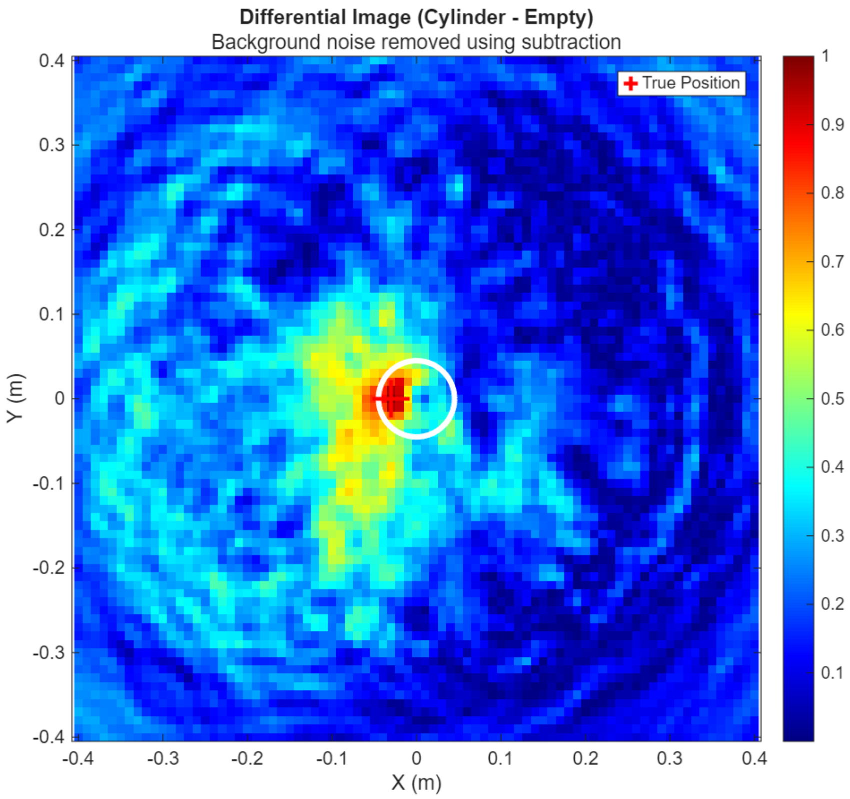

Figure 5: Differential Acoustic Image (Cylinder - Empty). The background noise is suppressed, revealing a high-contrast detection of the cylinder.

The imaging algorithm automatically detected the peak intensity at coordinates:

$$X = -0.03 \text{ m}, \quad Y = 0.00 \text{ m}$$

This suggests the cylinder was placed slightly off-center (3 cm to the left) during the experiment.

6 Error Analysis and Geometry Reconstruction

To validate the acoustic results, manual measurements of the distance between the microphones and the cylinder surface were recorded (Table 1).

Table 1: Manual measurements of distance from microphone to cylinder surface.

Mic Index	Dist (cm)	Mic Index	Dist (cm)
1	25.0	9	19.4
2	26.5	10	16.5
3	27.3	11	13.9
4	27.5	12	11.1
5	27.3	13	9.3
6	26.3	14	8.5
7	24.4	15	9.1
8	22.0	16	11.4

An optimization algorithm was implemented in MATLAB to fit the theoretical geometry to these manual measurements. The algorithm swept through possible cylinder coordinates and array orientations to minimize the error between the measured and calculated distances.

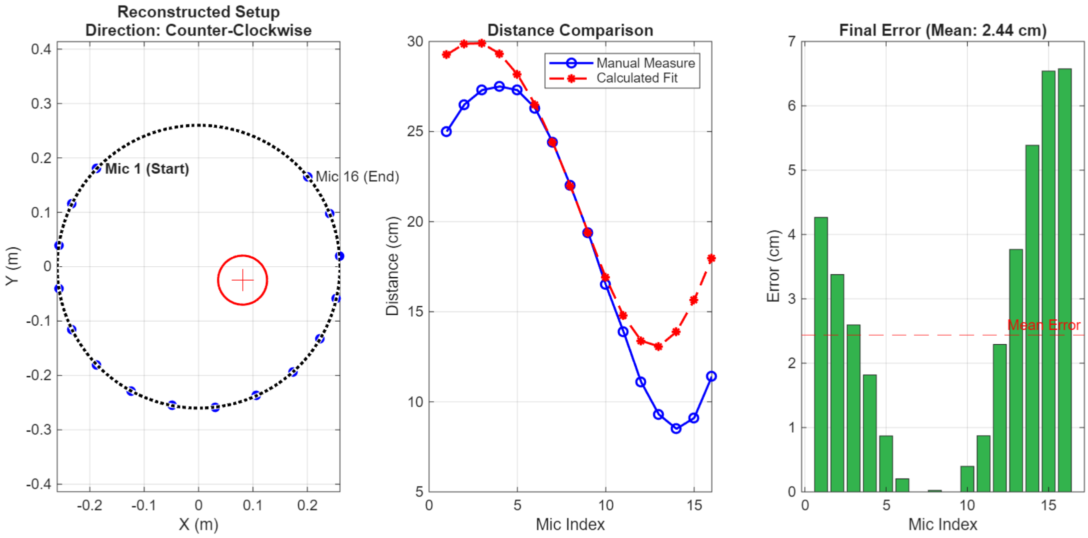

Figure 6: Reconstructed geometry based on manual measurements. Left: Top-down view showing the cylinder position relative to the arc. Center: Comparison of measured vs. fitted distances. Right: Error distribution.

The optimization converged with a **Mean Absolute Error of 2.44 cm**. The reconstructed geometry (Figure 6) confirms that the cylinder was located closer to microphones 13, 14, and 15, matching the acoustic image which showed the object shifted from the center. The consistency between the acoustic imaging result ($X = -3$ cm) and the manual geometry reconstruction validates the experimental approach.

7 Conclusion

This laboratory successfully demonstrated the capabilities of Passive Acoustic Imaging. By cross-correlating a diffuse noise field, we reconstructed the Green's functions between microphone pairs with high fidelity, evidenced by the clear time-domain pulses.

The application of Delay-and-Sum beamforming, particularly using the differential method, allowed for the precise localization of the scattering cylinder. The detected position ($X = -3$ cm) was cross-verified against manual measurements through a geometric optimization algorithm, yielding a low mean error of 2.44 cm. These results confirm that passive methods can effectively image an environment using only incoherent ambient noise.

8 Appendix: MATLAB Code

8.1 Pre-processing and Green's Function Retrieval

```
%% S. Rakotonarivo & Student
% PRE-PROCESSING FULL CODE
% Loads raw data -> Filters -> Calculates Green's Function ->
    Saves Result

clearvars; close all;

% --- PATH SETTINGS ---
Generaldir = 'D:\france semester\3_Acoustic Imaging\lab\
    Shahariar\DATA\';
FileDir = strcat(Generaldir, 'Data_MAT\');

% --- CHOOSE CONFIGURATION ---
FileDataName = 'MES_RealCylinder_';
FileSaveName = 'CC50_Cylinder.mat';

% Parameters
Nbmic = 16;
nReal = 50;
ii0 = 1;
fs_expected = 12.5e6 / 64 / 2;

% Filter Parameters
fmin = 1000; fmax = 9500;
ns_short = 1638; poverlap = 80;

% Load first file to check parameters
FileName = strcat(FileDir, FileDataName, num2str(ii0), '.mat');
load(FileName);
fs = A.fsample;
ns = length(A.t);

% Robust Butterworth Filter Design
[b, a] = butter(4, [fmin fmax]/(fs/2), 'bandpass');
data_filt = zeros(Nbmic*2, nReal, ns);

disp('Step 1: Filtering Data...');
for ii = ii0:1:ii0+nReal-1
    load(strcat(FileDir, FileDataName, num2str(ii), '.mat'));
    raw_signal = [A.yl; A.yr];
    for mic_idx = 1:(Nbmic*2)
        data_filt(mic_idx, ii, :) = filtfilt(b, a, raw_signal(
    mic_idx, :));
    end
end

disp('Step 2: Calculating Green''s Function...');
```

```
% CALLING THE STUDENT FUNCTION (Real output forced)
[C_t, duration_segment, n_segment] = FCT_Xcorr_Welch_Student(
    data_filt, ns_short, fs, poverlap);

% Derivative for Green's Function
dt = 1/fs;
dC_t = diff(C_t, [], 3);
Gt = dC_t(:, :, ns_short/2+1:end) ./ dt; % Causal part
t_Gt = 0:dt:(length(Gt)-1)*dt;

% Save
I.C_t = C_t; I.Gt = Gt; I.t_Gt = t_Gt;
I.fs = fs; I.Nbmic = Nbmic; I.c0 = 343;
save(strcat(FileDir, FileSaveName), 'I', '-v7.3');
```

Listing 1: PreProcessData_PassiveImaging_Student.m

8.2 Beamforming and Imaging

```
%% S. Rakotonarivo & Student
% FINAL REPORT GENERATOR (Differential Imaging)

clearvars; close all;
Generaldir = 'D:\france semester\3_Acoustic Imaging\lab\
    Shahariar\DATA\';
FileDir = strcat(Generaldir, 'Data_MAT\');
FileCyl = 'CC50_Cylinder.mat';
FileEmp = 'CC50_Empty.mat';

% Imaging Grid
step = 0.02; fov = 0.4;
x_vec = -fov:step:fov; y_vec = -fov:step:fov;
[GridX, GridY] = meshgrid(x_vec, y_vec);

% Load Data
load(strcat(FileDir, FileCyl)); I_Cyl = I;
load(strcat(FileDir, FileEmp)); I_Emp = I;

% Setup Mic Geometry
Nbmic = I_Cyl.Nbmic;
theta = linspace(0, 2*pi, Nbmic+1); theta(end) = [];
R_in = 0.26; R_out = 0.27;
XRec = [R_out * cos(theta), R_in * cos(theta)];
YRec = [R_out * sin(theta), R_in * sin(theta)];
c0 = 343; fs = I_Cyl.fs; t_Gt = I_Cyl.t_Gt;

% Beamforming Function
compute_image = @(Gt) run_das(Gt, fs, c0, XRec, YRec, t_Gt,
    x_vec, y_vec);

Img_Cyl = compute_image(I_Cyl.Gt);
```

```
Img_Emp = compute_image(I_Emp.Gt);

% Differential Imaging
Img_Diff = abs(Img_Cyl - Img_Emp);

% Detection
[~, idx] = max(Img_Diff(:));
[r, c] = ind2sub(size(Img_Diff), idx);
fprintf('DETECTED COORDINATES: X=%.2f m, Y=%.2f m\n', x_vec(c), y_vec(r));

% Plotting code omitted for brevity...

function Image = run_das(Gt, fs, c0, XRec, YRec, t_Gt, x_vec, y_vec)
    Image = zeros(length(y_vec), length(x_vec));
    for ix = 1:length(x_vec)
        for iy = 1:length(y_vec)
            d_all = sqrt((x_vec(ix) - XRec).^2 + (y_vec(iy) - YRec).^2);
            val = 0;
            for i = 1:32
                for j = 1:32
                    tau = (d_all(i) + d_all(j)) / c0;
                    idx = round(tau * fs) + 1;
                    if idx > 0 && idx < length(t_Gt)
                        val = val + abs(Gt(i, j, idx));
                    end
                end
            end
            Image(iy, ix) = val;
        end
    end
end
```

Listing 2: Final_Report_Generation.m

8.3 Geometry Optimization

```
%% S. Rakotonarivo & Student
% ADVANCED GEOMETRY OPTIMIZER

d_meas_cm = [25.0, 26.5, 27.3, 27.5, 27.3, 26.3, 24.4, 22.0, ...
             19.4, 16.5, 13.9, 11.1, 9.3, 8.5, 9.1, 11.4];
d_meas = d_meas_cm / 100;
radius_cyl = 0.045; R_mic = 0.26; spacing_deg = 17.56;

% Optimization Function
cost_ccw = @(v) calculate_error(v, d_meas, R_mic, radius_cyl, spacing_deg, 1);
```

```
cost_cw  = @(v) calculate_error(v, d_meas, R_mic, radius_cyl,
    spacing_deg, -1);

x0 = [0, 0, pi];
[res_ccw, err_ccw] = fminsearch(cost_ccw, x0);
[res_cw,  err_cw]  = fminsearch(cost_cw, x0);

if err_cw < err_ccw
    best_res = res_cw; direction = -1;
else
    best_res = res_ccw; direction = 1;
end

x_cyl = best_res(1); y_cyl = best_res(2);
fprintf('Mean Absolute Error: %.2f cm\n', mean(err_cw)*100);

function err = calculate_error(v, d_meas, R, r_cyl, sp_deg, dir
    )
    x = v(1); y = v(2); ang = v(3);
    th = ang + (0:15) * deg2rad(sp_deg) * dir;
    X_m = R * cos(th); Y_m = R * sin(th);
    d_model = sqrt((X_m - x).^2 + (Y_m - y).^2) - r_cyl;
    err = mean(abs(d_meas - d_model));
end
```

Listing 3: Geometry_Optimizer.m